A robot that acts to know

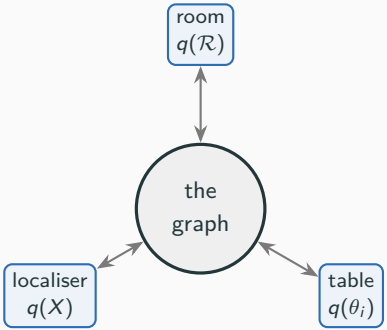
Day 3 — Belief in time

Course built on the doctoral thesis of Noé Zapata

Wednesday 2 December

Montevideo, 30 November – 4 December 2026

Yesterday: who holds a belief. Today: how long it lasts

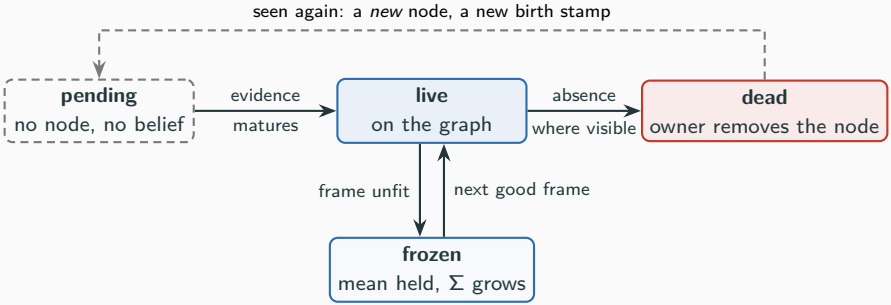


one F, divided among owners; no thresholds

Every node on that graph is a belief with a life of its own. Today:

- how it **begins**: model comparison, not counting;
- how it **ends**: absence as evidence;
- how it **fades**: forgetting as inference;
- what it **stands on**: a room, a pose, and an assignment of evidence;
- and what happens when all of that shares **one error**.

The life of an instance



Pending exists so that a wrong hypothesis costs nothing: nobody has read it yet.
Frozen suppresses an instance's *intake*, not the instance.

source: agents_life_cycle.tex l.379–423

How a belief begins

Birth is a comparison between two worlds

$$\Delta F = F^{(n+1)} - F^{(n)} = \underbrace{\mathbb{E}_q[-\ln p(\bar{Z} \mid \theta_{1:n+1})] - \mathbb{E}_q[-\ln p(\bar{Z} \mid \theta_{1:n})]}_{\text{accuracy gained by explaining those points}} + \underbrace{D_{\mathrm{KL}}[q(\theta_{n+1}) \,\|\, p(\theta_{n+1} \mid \ell)]}_{\text{complexity paid for the new object}}$$

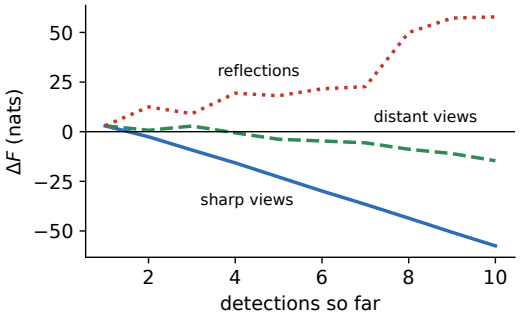


born exactly when the left side outweighs the right: $\Delta F < 0$

Two models of *different order*, one currency. The complexity is not a penalty added for regularisation: it is the same term every live instance already pays.

source: agents_life_cycle.tex l.594–622

A streak cannot read evidence



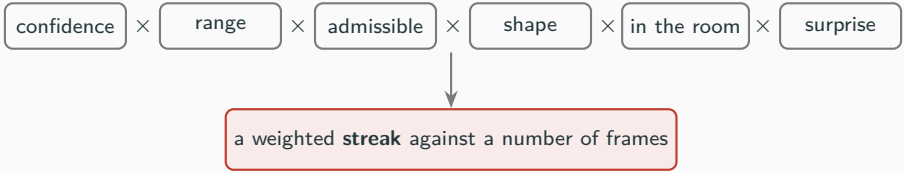
300 runs	streak of 3	$\Delta F < 0$
2 sharp views	0%	98%
10 reflections	94%	0%

A streak counts frames. ΔF reads what each frame says: how sharp it was, and how well the frames agree.

Two sharp views are worth more than ten reflections, and only nats can say so.

source: lab 3, exercises 1–2

What the robot actually computes: a threshold that should not be one



- Every factor stands in for one term of ΔF. As a decomposition it is reasonable.
- It loses the **units**: birth can be traded against nothing else in the model.
- Birth and death are decided in **different currencies**, so a whole subsystem exists to undo births that should not have happened.
- The proper comparison is within reach; what is missing is plumbing, not theory.

source: agents_life_cycle.tex l.629–706, l.1276–1282

The room is the first belief born

$$F(m) = \underbrace{\mathbb{E}_q[-\ln p(o \mid X, \theta_m, m)]}_{\text{inaccuracy}} + \underbrace{\kappa\,|V|}_{\text{complexity}}, \quad \kappa = \ln \frac{L}{\sigma}$$

The number of walls is unknown, so the room is a choice of *structure* m, priced like a birth. One coordinate to $\sigma = 3$ cm in a room 10 m across costs

$$\ln(10/0.03) \approx 5.8 \text{ nats.}$$

Minimum description length and minimum free energy are, here, one rule.



source: layout_acquisition.tex l.56–66, l.128–146

Two regimes, and a loop that is never formed

acquisition: the room is a variable



operating: the room is a constant

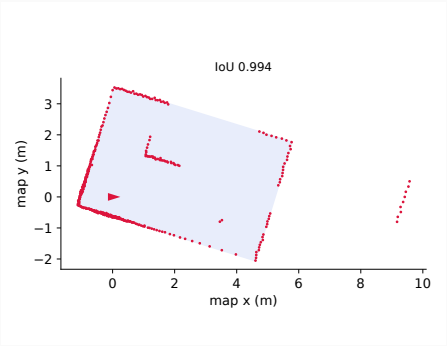


plain SLAM keeps running as a *shadow*:
a pose that measures how wrong the room might be

- A wrong wall would drag the pose, and the dragged pose would confirm the wrong wall.
- One factorised belief, two settings of precision. A room is a slow parameter; a pose is a fast state.
- The price: a constant cannot be doubted by the localiser that leans on it.

source: layout_acquisition.tex l.162–199

What tells a wall from a table: free space



In the lab, the table's south face earns **82 nats** as a candidate wall, against a price of 7.6. Evidence for a surface cannot keep furniture out of the room.

The grid therefore keeps two counts per 5 cm cell:

$$\text{matter}(c) \iff h_c > n_c$$

with h_c the beams that stop in the cell and n_c those that pass through it. Only the second count can say that a cell is *empty*.

Free floor beyond an edge is evidence against that edge.

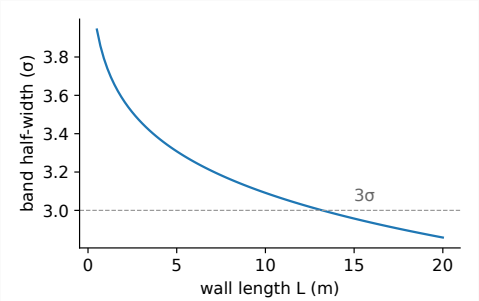
source: lab 3, exercise 11; layout_acquisition.tex l.507–526

A band that is derived, not chosen

Clutter is uniform over A. A wall puts its points in a strip L long, σ wide:

$$\frac{p_{\text{wall}}}{p_{\text{clutter}}} = \frac{A}{L\sqrt{2\pi}\,\sigma}\, e^{-r^2/2\sigma^2}$$

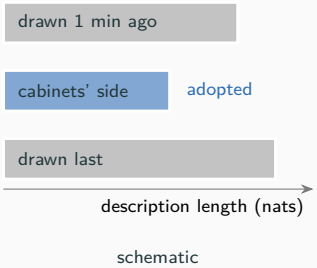
- Inlier iff the ratio exceeds 1: about 3σ.
- Each point brings ln(ratio) **nats**.
- The band **narrows as L grows**: a long wall is a more expensive explanation.



A tuned threshold gives the band. The derivation gives the band *and* the currency.

source: lab 3, exercise 8; the same mixture, per cell: layout_acquisition.tex l.1049–1055

Promotion: the outline with the shortest description



schematic

The outline moves between wall and furniture as the robot drives, so "the latest one" would make the room depend on when exploration stopped. Up to 60 recent outlines are kept and scored on *all* the evidence.

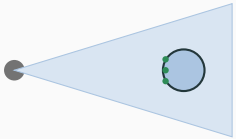
- In 7 of 10 apartment runs (earlier build) the winner was not the last drawn.
- From the kitchen-corner start: overlap 0.968 at promotion, 0.983 refined.
- Ten generated rooms: median 0.994.

All in simulation; the pipeline has not run on the physical robot.

source: layout_acquisition.tex l.1036–1046, l.1173–1177, l.1455–1462

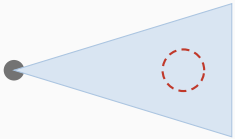
How a belief ends

A miss is evidence only where the object would have been seen



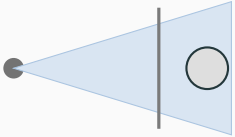
occupancy: a return

confirms



predicted visible, absent

removes



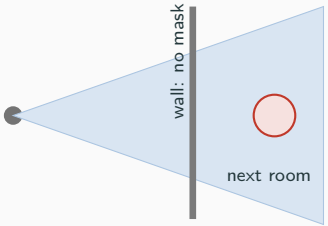
occluded or out of view

holds

The whole stage lies between the middle panel and the right one. Existence is one log-odds L per instance, clamped to $\pm L_{\max}$, so it stays finite and recoverable.

source: agents_life_cycle.tex l.887–919

A factor whose argument list was wrong



projects into view, lands on unmasked pixels, finds no mask, votes for its own removal

$$P(M \mid \theta, \ell, X) \longrightarrow P(M \mid \theta, \ell, X, \mathcal{R})$$

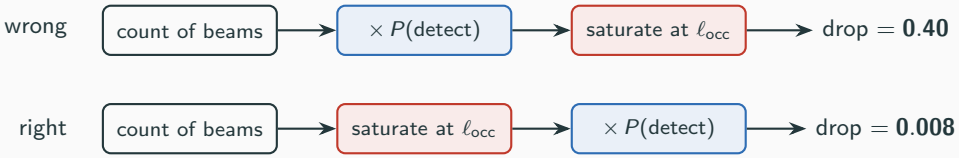
A wall is not a detector class, so without the room polygon every wall is transparent.

Measured: the same defect, three agents

Refrigerator, then door, then table and cabinet. Each copy was consistent with itself, so no per-agent test could find it.

source: agents_life_cycle.tex l.921–951; CONCEPT_AGENT_LIFECYCLE.md l.7–11

Weight absence by $P(\text{detect})$, on the correct side of the ceiling

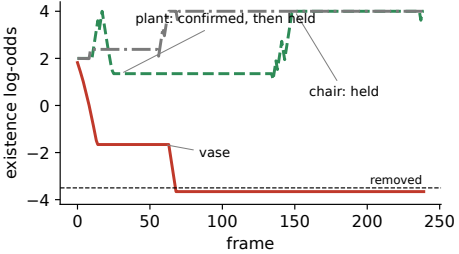


A view in which a present object would be only 2% resolvable.

A modulation applied upstream of a saturation is invisible: the saturation absorbs it.

source: agents_life_cycle.tex l.953–973; numbers: lab 3, exercise 4

Four rules, one tour



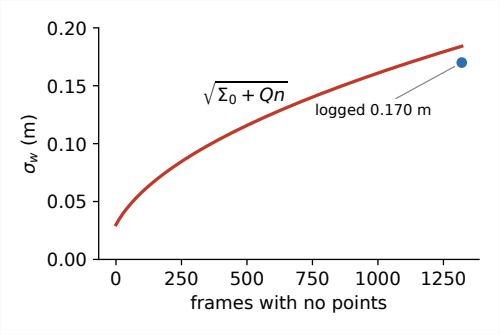
rule	removes (frame)
evidence	vase (68)
no walls in line of sight	vase (68), chair (174)
weight the count	vase (13)
miss counter	everything by frame 44

Never resolvable? No decay constant. The instance is routed to a **verification pull**: uncertainty that cannot be resolved from here is a reason to move.

source: lab 3, exercise 4; agents_life_cycle.tex l.975–985

How a belief fades

A constant Q forgets on the agent's clock



$\Sigma \leftarrow \Sigma + Q$, every frame, seen or not, with $Q = (0.005\ \text{m})^2$.

- A converged table ($\sigma_w \approx 3$ cm) loses half its precision in **3.6 s**.
- It is back at the prior after about **5.9 minutes**, whether it was tracked for thirty seconds or for three hours.

Measured: a real table, 1320 blind cycles

Predicted $\sigma_w = 0.184$ m; logged 0.170 m. The next sparse mask re-cut the table.

source: agents_life_cycle.tex l.1007–1043

Observed, precision stops at a floor

One predict plus one update, $\Sigma^{-1} \leftarrow \Sigma^{-1} + i$, with y the predicted precision and $\varrho = Q$:

$$\frac{1}{y} = \frac{1}{y+i} + \varrho \quad \Longrightarrow \quad \boxed{\sigma_{\text{steady}} \approx (\varrho / i)^{1/4}}$$

- Elapsed time does not appear. Evidence is discarded at the rate it is acquired: three hours of observation leave the belief where thirty seconds did.
- Lab: formula 10.0 mm, simulated 9.4 mm.
- Clamping Σ caps the first problem and does nothing about this one.

Nothing an agent learns extends how long it stays resistant.

source: agents_life_cycle.tex l.1045–1082; lab 3, exercise 5

A constant becomes an inferred quantity

Each instance carries a log-volatility, $Q = e^{\zeta}$, driven by its own surprise:

$$\zeta \leftarrow \zeta + \eta\Big(\underbrace{\tfrac{1}{2}\,\tilde{d} - \tfrac{1}{2}\mathbb{E}[\tilde{d}\,]}_{\text{surprise seen minus expected}} - \underbrace{\frac{\zeta - \zeta_0}{\sigma_\zeta^2}}_{\text{hyperprior pull-back}} \Big)$$

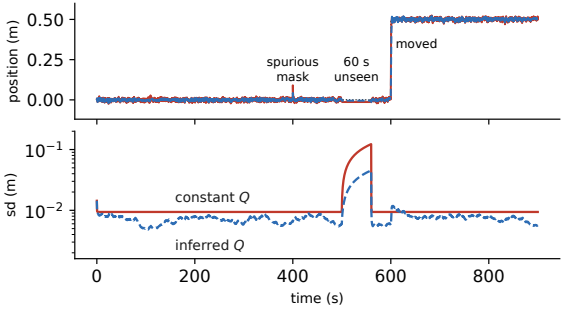
- Consistent views: ζ falls, memory lengthens, precision accumulates.
- A sustained surprise: ζ rises and the model adapts. Stiffness is never a lock.

In the lab $\tilde{d}$ is the saturating (Student-t) surprise of the innovation: one outrageous frame moves ζ by a bounded amount.

Design only: no agent runs this yet.

source: agents_life_cycle.tex l.1085–1125 (form of the update); lab 3, exercise 5

In the toy: stiff, but not locked



	constant	inferred
sd before (mm)	9.4	6.5
spurious mask (mm)	92	44
after 60 s unseen (mm)	123	45
frames to adapt	9	15

The failure mode: a steady **bias** looks like agreement. A 5 cm bias ends 5.4σ out with constant Q, 7.6σ with inferred.

source: lab 3, exercises 5–6; agents_life_cycle.tex l.1127–1133

Localisation is one more belief

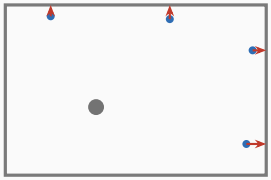
The room supplies the likelihood

For a scan $\{p_i\}$ and a candidate pose X:

$$d_i(X) = \operatorname{dist}\left(R(\phi)p_i + t,\ \mathcal{R}\right)$$

$$\mathcal{L}_{\text{obs}} = \frac{1}{2\sigma^2} \cdot \frac{1}{N} \sum_i w_i \operatorname{Huber}(d_i)$$

- The residual *is* the prediction error: no expected scan is rendered.
- Note the $\frac{1}{N}$: the precision does not grow with scan density.
- Range and incidence weights are precisions on a covariate, not gates.



each return is pulled onto its nearest wall

source: room_concept.tex l.243–307

What the residual floor is made of



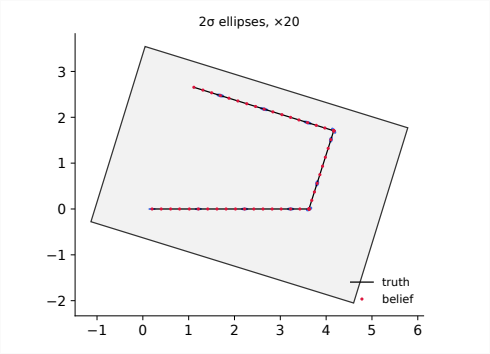
Park the robot. The mean wall distance of each scan hovers near 87 mm and never reaches zero.

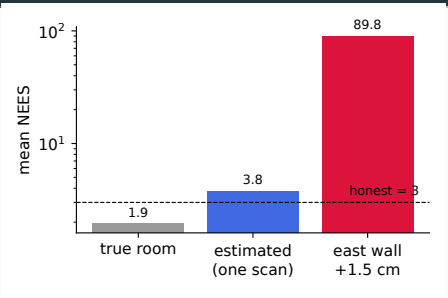
- Its fluctuation is white: $N^{-1/2}$, measured ratio 1.41 against a predicted $\sqrt{405/200} = 1.42$.
- That fixes the per-point scatter at ≈ 37 mm, which explains $0.798 \times 37 \approx 30$ mm.

Two-thirds of the floor is an error every return shares. More returns cannot remove it.

source: room_concept.tex l.386–426

Localising against the room you just built



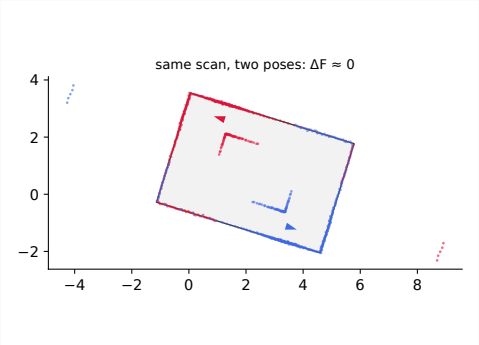


A wall **1.5 cm** out of place makes the localiser confidently wrong: NEES 90 where 3 is honest.

In the operating regime the polygon is a constant, so its error is shared by every scan, for ever.

source: lab 3, exercise 13

Symmetry: where descent cannot help



A rectangle looks the same after a half-turn: the same scan fits both poses with *identical* free energy, and no downhill path joins them.

They are separate hypotheses, so the agent enumerates them and chooses: a discrete search, then local refinement.

This is one of the few places where a threshold is right: a threshold is defensible exactly when the alternative cannot be expressed as a covariance.

source: room_concept.tex l.1869–1927

Whose evidence is it?

Association: the factor the free energy cannot see



a poorly known instance accepts detections from further away

$$P(M \mid X, Z, \Phi) \approx \mathbf{1}[M = \hat{M}], \quad S = \Sigma^{xy}_{\theta_i} + \sigma^2_{\text{det}} I$$

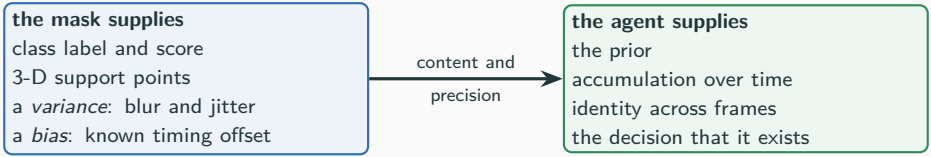
Squared Mahalanobis distance at most 9, assigned greedily, lowest cost first.

A distribution on one value has no entropy and no expectation: the matching is free, in the literal sense that F cannot see it. A wrong match is invisible to the quantity that would have registered it.

The repair comes later and from elsewhere: merging, and the existence belief.

source: concept_agents.tex l.483–486, l.859–889

A mask is a qualified likelihood



- The producer reports the physics of capture; the consumer decides what that does to its belief. The hub gates nothing.
- Variance and bias are published separately because they are different: symmetric error downweights, biased error gates.
- The qualification is a prior on the *channel*, not on the world.

source: phylogenetic_perception.tex l.347–408

Shared error

The common-mode ceiling

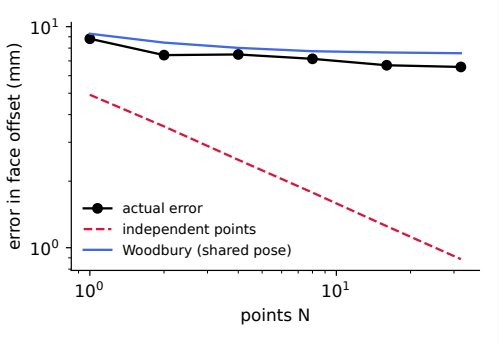
N residuals that share one offset $\delta \sim \mathcal{N}(0, \Sigma_\delta)$, a pose error, plus independent noise. Marginalise δ:

$$\mathcal{I} = J_z^\top \left(\sigma^2 I + J_z \Sigma_\delta J_z^\top\right)^{-1} J_z \overset{\text{Woodbury}}{=} \left(\sigma^2 (J_z^\top J_z)^{-1} + \Sigma_\delta\right)^{-1} \xrightarrow[N\to\infty]{} \Sigma_\delta^{-1}$$

- A denser mask eventually stops helping. The ceiling is set by the *pose* error, not by the pixel count.
- The joint information on (object, offset) has rank 6, not 12: the data cannot tell a displaced object from a displaced robot.

source: concept_agents.tex l.1048–1090

The ceiling, measured in the toy



The robot maps the table's face from a *dead-reckoned* pose.

- Treating points as independent claims under 1 mm at $N = 32$.
- The real error stalls near 7 mm.
- Woodbury predicts it to within about 15% at every N.

Averaging removes independent error. It does not remove shared error.

source: lab 3, exercise 15

Case study: association dies by a common-mode latch

work in progress

One frame of a real recording, on an earlier estimator:

seg	χ^2	$\Delta\phi$	gate σ_ϕ
0	1.03	-5.85°	6.78°
1	1.02	-5.76°	6.34°
2	1.20	-5.79°	6.01°
3	1.06	-5.77°	6.67°

"A parked run looks perfect and the first tour kills it."

The *same* -5.8° on every wall is one **heading** error, not four wall errors.

Each wall is gated independently against an error they all share. When it crosses $\approx 6^\circ$ they all fail in the same frame, nothing is left to correct the pose, and association goes from 100% to 0% for ever.

source: post-thesis note, 2026-09-18

Lab

Lab 3: a belief's life, and what it stands on

`labs/day3.ipynb`, about two hours.

How a belief begins and ends

A. **Birth**: ΔF against a streak of three.

B. **Absence**: line of sight through walls; the weight after the ceiling; four rules on one tour.

C. **Forgetting**: constant Q against inferred volatility; then a bias.

What it stands on

D. **The room**: the vote, the band, four walls; does the table pay the price?

E. **The pose**: drive the route; NEES on three maps.

F. **Symmetry**: F at the true pose and its half-turn.

G. **Ceiling**: independent against Woodbury.

Short of time: A, B, C, E and G.

Tomorrow

Today the robot watched. Tomorrow it decides:

- objects publish what they can do for its uncertainty, priced in nats;
- expected free energy chooses, and where to look next is experiment design;
- routes and tracking become control problems with no separate obstacle term;
- and it turns the same objective on itself: calibration as inference.

Reading for Day 4

Thesis chapters:

- *Action: affordances and the expected free energy*, up to "Why the gradient decomposes";
- *Self-Calibration of the Robot's Body*: "What the body is", "What no data can determine", "What each motion makes observable".